

From Field to Market: How Corn Yields Affect Economic Stability in the State of Ohio and
Clinton County

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Abstract

This study examines the relationship between corn yield fluctuations and key socioeconomic indicators, specifically unemployment rates and Supplemental Nutrition Assistance Program (SNAP) participation, at both the state level (Ohio) and the county level (Clinton County, Ohio). Using publicly available annual data from the United States Department of Agriculture National Agricultural Statistics Service (USDA NASS) and the Federal Reserve Economic Data (FRED), the study applies correlation analysis and linear regression to evaluate the strength and direction of these relationships.

The results show a clear difference based on geographic scale. At the state level, higher corn yields are significantly associated with lower unemployment ($p = 0.030$). However, higher yields are also associated with increased SNAP enrollment ($p < 0.000$), suggesting that broader socioeconomic factors such as urbanization may influence SNAP participation more strongly than agricultural productivity alone.

At the local level in Clinton County, higher yields are also associated with decreased SNAP enrollment ($p < 0.001$), indicating that in agricultural communities, productive harvests may play a direct role in reducing food insecurity. The relationship between local yields and unemployment is positive but not statistically significant ($p = 0.063$), indicating a weak association that falls just outside conventional significance thresholds. However, the result is still notable as it is close to the 0.05 cutoff, suggesting a potentially meaningful trend that may warrant further investigation with additional data or a larger sample size.

Overall, these findings suggest that the economic effects of agriculture are more directly observable at the local level than at broader state scales, where additional structural factors influence socioeconomic outcomes.

I. Introduction

When a drought stunts corn growth across the fields of Clinton County, Ohio, the consequences do not end at the farm gate, they ripple outward, touching the livelihoods of farm laborers, the coffers of local businesses, and the social safety nets that thousands of families depend on to put food on their tables. Agriculture is a cornerstone of the American economy, contributing approximately 5.5 to 5.6 percent of the nation's gross domestic product and supporting well over 21 million jobs across the supply chain (USDA Economic Research Service, 2023). In rural states such as Ohio, this contribution is even more pronounced. Ohio consistently ranks among the top corn-producing states in the nation, with corn constituting between 19 and 24 percent of the state's total agricultural output by value. Clinton County, a largely rural county in southwestern Ohio, exemplifies the kind of community where agricultural production and local economic welfare are deeply intertwined. In such communities, the health of the harvest season directly shapes income

levels, seasonal employment opportunities, and households' ability to meet basic needs without federal assistance.

Corn is not only the dominant cash crop in Ohio but also a commodity whose yield is subject to significant year-to-year variability driven by weather events, input costs, pest pressures, and commodity market forces. A poor corn yield season can reduce farm revenues, limit demand for agricultural labor, and suppress the economic activity of businesses like feed stores, equipment dealers, and processing facilities. Conversely, a bumper crop year tends to stabilize farm incomes and reduce reliance on public assistance programs. Understanding these dynamics is essential for crafting effective agricultural and social policy.

Despite the intuitive connection between agricultural output and rural economic well-being, empirical studies that quantify the relationship between specific crop yields and socioeconomic safety-net indicators at the county scale are relatively scarce. Most existing research either focuses on national-level aggregate trends or examines the effects of broad agricultural shocks rather than the annual variability in a single commodity's yield. This project addresses that gap by examining how fluctuations in corn yields in Ohio and Clinton County correlate with unemployment rates and SNAP benefit recipient counts over the 2001–2024 period.

This paper argues that fluctuations in corn yields in Ohio and Clinton County are meaningfully associated with changes in unemployment rates and SNAP benefit reciprocity, and that the nature and magnitude of these associations differ between the state and county scales, with Clinton County exhibiting a stronger agricultural-economic linkage due to its rural character and concentrated dependence on corn production.

II. Literature Review

The relationship between agricultural productivity and rural economic indicators has been explored across a variety of disciplinary frameworks. Economists have long acknowledged that agricultural shocks, defined broadly as unexpected deviations from expected crop output — can transmit through rural labor markets and public assistance utilization. Gundersen and Oliveira (2001) demonstrated that farm-level income volatility has measurable effects on food insecurity rates in rural counties, establishing an empirical basis for connecting crop performance to nutrition assistance demand. Similarly, research by Mishra and El-Osta (2002) found that income variability in farm households significantly predicted participation in government transfer programs, including food assistance.

More recently, scholars have examined the role of specific commodity markets. Work by Barnett and Coble (2012) on crop insurance uptake in the Corn Belt implicitly highlighted the economic vulnerability of corn-dependent communities to yield shocks. At the state level, studies of Ohio's agricultural sector by the Ohio State University Extension Service have documented the multiplier effects of corn production, estimating that each dollar of farm gate corn revenue generates between 1.5 and 2.2 dollars in broader economic activity. This multiplier dynamic suggests that yield declines could have amplified downstream effects on employment and household food security.

The SNAP program itself has been the subject of substantial economic research. Bitler and Hoynes (2010) found strong countercyclical patterns in SNAP enrollment, with participation rising sharply during economic downturns and declining in periods of growth. While their analysis focused primarily on macroeconomic business cycles, their framework provides a useful lens through which agricultural cycles, which can be partially decoupled from broader economic trends, can also be examined as drivers of SNAP enrollment.

Unemployment in rural communities has similarly been studied in relation to farm income. Goetz and Swaminathan (2006) found that counties with higher concentrations of farm employment exhibited greater unemployment volatility in response to agricultural income shocks, a finding particularly relevant to Clinton County's relatively farm-dependent labor market.

This study is grounded in two complementary theoretical frameworks. The first is the agricultural income-transmission model, which posits that changes in farm revenues flow through local economies via labor demand, business revenues, and household income, ultimately influencing access to and need for social safety-net programs. When corn yields fall, farm operators earn less, reduce spending on inputs and hired labor, and may draw on or trigger eligibility for federal assistance programs. Conversely, high-yield years boost farm incomes, sustain or grow seasonal labor demand, and reduce households' need for SNAP or other supports.

The second framework is the economic base theory of regional development, which identifies a community's export-oriented economic activities, in Clinton County's case, corn and other commodity crops, as the primary engine of local income generation. Under this theory, shocks to the base sector (agriculture) have multiplied effects on the service and retail sectors, ultimately reshaping employment patterns and income levels across the whole community. Together, these frameworks predict that corn yield variability should be negatively correlated with unemployment (lower yields, higher unemployment) and positively correlated with SNAP enrollment (lower yields, more SNAP recipients), with the effect being more pronounced in a rural, agriculture-dependent county such as Clinton than in the broader, more economically diversified state of Ohio.

Several important knowledge gaps remain in the existing literature. First, most studies examining agricultural-economic linkages use national or regional data, leaving county-level

dynamics underexplored. Rural counties can exhibit economic behaviors that diverge substantially from state or national averages, yet policy is rarely calibrated at this granular level. Second, while SNAP and unemployment are widely studied individually, their joint relationship with a specific commodity's yield, particularly at annual resolution over multi-decade periods, has received limited attention. Third, no published study appears to have directly compared how these agricultural-economic linkages differ between Ohio as a whole and a representative rural Ohio county, making this project's comparative dimension a novel contribution. Finally, the role of confounding factors, particularly the economic disruptions of the 2008–2009 financial crisis, the COVID-19 pandemic (2020–2021), and drought events, in mediating these relationships has not been systematically accounted for in prior county-level research.

III. Methodology

All data used in this analysis were obtained from publicly available government sources, including the United States Department of Agriculture National Agricultural Statistics Service (USDA NASS) for corn yield data and the Federal Reserve Economic Data (FRED) for unemployment and SNAP-related indicators. The study period covers 2001 to 2024, with all datasets standardized to a consistent monthly time frame ending in June 2024 to allow for direct comparison across variables.

Due to inconsistencies in format and frequency across datasets, substantial data preprocessing was required prior to analysis. Date variables were standardized across all datasets into a consistent structure using pandas datetime functions, with mixed formats such as “YYYY-MM-DD” and “Month Year” converted into a unified monthly representation. This ensured that all observations could be aligned temporally. Missing values and inconsistencies were handled

using dataset-specific approaches. For example, in the Clinton County SNAP dataset, missing 2024 values were imputed using the mean of existing observations to preserve continuity in the time series. Duplicate observations were removed to ensure that each time period contained a single valid entry.

Several datasets originally reported at annual frequency were expanded to monthly frequency using row repetition in order to match the resolution of SNAP and unemployment data. After expansion, datasets were trimmed to ensure alignment within the study period and to remove any excess or future observations beyond June 2024. All datasets were then synchronized using a common monthly date index derived from the SNAP dataset, ensuring consistent temporal alignment across unemployment, SNAP, and corn yield. Population data for both Ohio and Clinton County were similarly expanded from annual to monthly frequency and aligned to the same index. Excess future estimates outside the study period were removed to avoid distortion in the analysis.

Corn yield data required additional processing due to multiple reporting categories and forecast-based observations. Only grain corn yield values were retained, while silage-related entries were excluded. Forecast periods were systematically ordered using a predefined mapping of reporting months to ensure correct sequencing of observations. In cases where data were missing, such as the October 2013 reporting gap caused by a government shutdown, values were imputed using adjacent forecast data to maintain continuity. The resulting agricultural dataset was then expanded and aligned to the same monthly structure as the socioeconomic variables.

After preprocessing, all datasets were merged into integrated state-level and county-level datasets containing unemployment rates, SNAP participation, corn yields, and population

variables. These were then combined into a unified dataset to enable comparative analysis across geographic scales. All final datasets were exported as CSV files to ensure reproducibility and transparency.

The analysis began with exploratory visualization of each variable to examine long-term trends in unemployment, SNAP participation, corn yields, and population for both Ohio and Clinton County. These time-series plots were used to identify structural differences and general movement patterns across the study period. Following this, Pearson correlation coefficients were computed to measure the strength and direction of relationships between corn yields and socioeconomic indicators. Correlations were calculated separately for Ohio and Clinton County to allow for geographic comparison.

To further explore these relationships, simple linear regression models were applied and visualized using scatter plots with fitted trend lines. These models provided a visual and statistical representation of the direction and magnitude of associations between variables. To account for potential lagged effects between agricultural production and socioeconomic outcomes, a time-shifted dataset was also used in parts of the analysis to better reflect delayed economic responses.

Statistical significance was evaluated using p-values from Pearson correlation tests, with a conventional threshold of $p < 0.05$ used to determine significance. Results were interpreted in both statistical and practical terms, particularly in cases where values approached but did not meet the threshold for significance. All analysis, data processing, and visualization were conducted in Python using pandas, NumPy, Matplotlib, and SciPy.

IV. Results and Findings

The analysis reveals distinct relationships between corn yields and socioeconomic indicators, with notable variation between the state and county levels.

At the state level (Ohio), corn yield exhibits a statistically significant negative relationship with unemployment rates ($p = 0.030$). This suggests that higher agricultural productivity is associated with improved labor market conditions. Years with strong corn yields likely correspond to increased farm income, greater demand for agricultural labor, and broader economic spillover effects that stabilize employment.

In contrast, corn yield shows a positive and statistically significant relationship with SNAP participation at the state level ($p = 0.00$). While this result appears counterintuitive, it reflects the complexity of statewide socioeconomic dynamics. Ohio's population includes large urban centers where SNAP participation is influenced more by structural economic conditions than by agricultural performance. Thus, increases in corn yield do not necessarily translate into reduced reliance on food assistance at the aggregate state level.

At the county level (Clinton County), the relationships are more directly aligned with theoretical expectations. Corn yield is negatively and significantly associated with SNAP participation ($p = 0.000$), indicating that higher yields correspond with reduced food assistance usage. This supports the idea that in rural, agriculture-dependent communities, productive harvests directly enhance household financial stability and food security.

The relationship between corn yield and unemployment in Clinton County is positive but not statistically significant ($p = 0.063$). Although this does not meet the conventional threshold for significance, the proximity to 0.05 suggests a potential underlying relationship. This may indicate that local labor markets are influenced by additional factors such as seasonal

employment patterns, mechanization, or labor substitution effects that weaken the direct link between yield and employment.

V. Discussion

The findings of this study highlight the importance of geographic scale in understanding the economic impact of agricultural productivity. While theory predicts that higher agricultural output should reduce unemployment and reliance on assistance programs, this relationship holds more consistently at the local level than at the state level.

In Clinton County, the results strongly support the agricultural income-transmission model. Higher corn yields are associated with reduced SNAP participation, indicating that agricultural success directly improves household well-being. This reflects the county's economic structure, where agriculture plays a central role in income generation and employment. The findings suggest that policies aimed at stabilizing or improving agricultural productivity could have meaningful impacts on reducing food insecurity in rural communities.

In contrast, the state-level results illustrate the limitations of applying localized economic theories to broader, more diverse populations. The positive relationship between corn yield and SNAP participation in Ohio suggests that agricultural productivity is only one of many factors influencing economic stability. Urbanization, wage inequality, and macroeconomic conditions likely exert stronger influence on SNAP enrollment than agricultural output alone.

The near-significant relationship between corn yield and unemployment in Clinton County warrants further investigation. It is possible that technological advancements in

agriculture have reduced the need for labor even during high-yield years, weakening the expected negative relationship.

This study contributes to the literature by providing a direct comparison between state and county-level agricultural-economic linkages, demonstrating that aggregation can mask important local dynamics. The results emphasize the need for policy approaches that are tailored to regional economic structures, rather than relying solely on statewide or national trends.

Several limitations should be acknowledged. First, the analysis does not fully account for external shocks such as the 2008 financial crisis or the COVID-19 pandemic, which may have influenced both agricultural production and socioeconomic indicators. Second, the use of linear models may oversimplify relationships that are potentially nonlinear. Future research could incorporate more advanced modeling techniques.

In conclusion, this study demonstrates that agricultural productivity plays a meaningful role in shaping economic outcomes, particularly in rural communities, but its effects are mediated by broader structural factors at higher levels of aggregation. Understanding these differences is critical for designing effective economic and agricultural policies.

References

- Barnett, B. J., & Coble, K. H. (2012). Understanding regional differences in farm risk and why it matters for farm policy. *American Journal of Agricultural Economics*, 94(2), 361–368.
- Bitler, M. P., & Hoynes, H. W. (2010). The state of the safety net in the post-welfare reform era. *Brookings Papers on Economic Activity*, 2010(2), 71–127.

- Bureau of Labor Statistics. (2024). Local area unemployment statistics. U.S. Department of Labor.
<https://www.bls.gov/lau/>
- Goetz, S. J., & Swaminathan, H. (2006). Wal-Mart and county-wide poverty. *Social Science Quarterly*, 87(2), 211–226.
- Gundersen, C., & Oliveira, V. (2001). The Food Stamp Program and food insufficiency. *American Journal of Agricultural Economics*, 83(4), 875–887.
- Mishra, A. K., & El-Osta, H. S. (2002). Risk management through enterprise diversification: A farm-level analysis. *Agricultural Finance Review*, 62(2), 151–164.
- Ohio State University Extension Service. (2022). Ohio agricultural statistics: Corn production and economic impact. The Ohio State University College of Food, Agricultural, and Environmental Sciences.
- U.S. Department of Agriculture Economic Research Service. (2023). Ag and food sectors and the economy. USDA ERS. <https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/>
- U.S. Department of Agriculture National Agricultural Statistics Service. (2024). Quick Stats agricultural database. USDA NASS. <https://quickstats.nass.usda.gov/>
- U.S. Department of Agriculture Food and Nutrition Service. (2024). Supplemental Nutrition Assistance Program (SNAP) data. Federal Reserve Bank of St. Louis FRED. <https://fred.stlouisfed.org/>